

leftward force on Y. Adding up, total magnitude will be  $3 F$

**23. Ans: A**

Explanation: The soft iron core increases the flux linkage through coil Q. Hence the rate of change of flux linkage is larger, resulting in a larger induced e.m.f. in coil Q. However, the frequency of the signal remains unchanged as the frequency of the alternating voltage in P remains unchanged.

**24. Ans: A**